

Classification of Crocs Purchase Intention Using Artificial Neural Networks with Sigmoid and ReLU Activations

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ABSTRACT

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Crocs has experienced a strong resurgence in popularity among Indonesian young consumers, yet the factors that drive their purchase intention remain insufficiently understood. This study classifies the purchase intention of Crocs footwear using an Artificial Neural Network (ANN) with two activation functions, such as sigmoid and ReLU. Data were collected through an online questionnaire distributed to 300 respondents from Prasetiya Mulya University, measuring five predictor constructs—Price, Promotion, Quality, Customer Review, and Brand Awareness—and the target construct Purchase Intention on a 1–5 Likert scale. The purchase intention score was binarized into Low (<4) and High (≥ 4) classes. Both models were trained on standardized features using a (16, 8) hidden-layer architecture and evaluated via 5-fold cross-validation. The ANN-Sigmoid model achieved a mean cross-validation accuracy of 81.0% and an F1-score of 0.78, outperforming the ANN-ReLU model (78.0% accuracy, 0.73 F1-score). Permutation feature importance on the best model identified Quality, Customer Review, and Price as the strongest contributors to predicting Crocs purchase intention, while Brand Awareness had a negligible effect. The findings suggest that Crocs' marketing strategy in Indonesia should emphasize tangible product attributes rather than rely on brand recognition alone.

1. Introduction

The footwear industry has become an increasingly competitive segment of the global consumer goods market, with brands constantly innovating in product design, marketing, and brand positioning to capture young consumers. Crocs, introduced in 2002 as a functional clog made from a proprietary closed-cell resin called Croslite, has evolved from a niche utilitarian shoe into a globally recognized lifestyle brand. After years of fluctuating perception, Crocs has experienced a strong revival in popularity, particularly among Gen Z and younger millennials, driven by celebrity collaborations, social media trends, and the rise of comfort-oriented fashion [1].

In Indonesia, Crocs has gained significant traction among university students, where it is widely worn as casual footwear for daily campus mobility. Despite this visible popularity, the underlying factors that shape consumer purchase intention toward Crocs in the Indonesian student segment have not been extensively studied. Understanding purchase intention is critical for marketers, since intention is widely recognized as the strongest behavioral predictor of actual purchase decisions [2].

Previous research on purchase intention has identified several recurring antecedents, including perceived price, promotional activity, product quality, consumer reviews, and brand awareness. However, these factors often interact in non-linear ways that are difficult to capture using traditional linear models such as multiple regression. To address this limitation, machine learning approaches—particularly Artificial Neural Networks (ANN)—offer a flexible alternative capable of modeling complex non-linear relationships between predictors and consumer behavior [3].

Within ANN, the choice of activation function plays a central role in determining model performance. The sigmoid (logistic) function maps inputs to a bounded range and is widely used in binary classification problems, while the Rectified Linear Unit (ReLU) has become popular in modern deep learning due to its computational efficiency and ability to mitigate the vanishing-gradient problem [4]. Comparing these two activation functions in the context of consumer behavior modeling provides practical insight into which configuration is more suitable for small-to-medium-sized survey data.

This study aims to classify the purchase intention of Crocs footwear among university students using an ANN model with two different activation functions (sigmoid and ReLU), based on five predictor constructs: Price (X1), Promotion (X2), Quality (X3), Customer Review (X4), and Brand Awareness (X5). The contribution of this research is twofold: (1) it provides empirical evidence on the drivers of Crocs purchase intention in Indonesia, and (2) it benchmarks two activation functions on the same dataset to inform future ANN-based consumer behavior studies.

2. The Proposed Method

2.1. Research Framework

The proposed method follows a supervised classification pipeline. Five Likert-scale constructs—Price (X1), Promotion (X2), Quality (X3), Customer Review (X4), and Brand Awareness (X5)—are used as input features to predict whether a respondent has a High or Low Purchase Intention toward Crocs. Each construct is measured using three indicator items adapted from prior consumer behavior literature, and the construct score is computed as the arithmetic mean of its indicators.

- The classification task is formulated as a binary problem: respondents with a mean Purchase Intention score of 4 or higher are labeled as High Intention (class 1), while those with a score below 4 are labeled as Low Intention (class 0). The threshold of 4 corresponds to the "agree" point on the 5-point Likert scale and is consistent with prior studies that treat this level as the minimum indication of positive intention.

2.2. ANN Architecture and Activation Functions

Two Artificial Neural Network classifiers are constructed using the same architecture: an input layer of 5 neurons (one per predictor), two hidden layers with 16 and 8 neurons, and a single output neuron for the binary class. The two models differ only in the activation function applied to the

hidden layers, namely sigmoid (logistic) and ReLU. The two models differ only in the activation function applied to the hidden layers, namely sigmoid (logistic) and ReLU.

For a given hidden neuron, the pre-activation is computed as a weighted sum of its inputs plus a bias term, as expressed below :

$$z_j^{(l)} = \sum_{i=1}^n w_{ji}^{(l)} a_i^{(l-1)} + b_j^{(l)}$$

where $z_j^{(l)}$ is the pre-activation of the j -th neuron in layer l . $w_{ji}^{(l)}$ is the weight connecting the i -th neuron in layer $l-1$ to the j -th neuron in layer l , $a_i^{(l-1)}$ is the output of the previous layer, and $b_j^{(l)}$ is the bias term. The pre-activation is then transformed by an activation function $f()$ to produce the neuron's output $a_j^{(l)} = f(z_j^{(l)})$.

The first model uses the sigmoid (logistic) activation function, defined as below :

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

The sigmoid function maps any real-valued input into the bounded range (0, 1), which makes it well-suited for probabilistic interpretation in binary classification tasks.

The second model uses the Rectified Linear Unit or ReLU activation, defined as below :

$$ReLU(z) = \max(0, z)$$

ReLU outputs zero for any negative input and passes positive inputs unchanged. It is computationally efficient and helps mitigate the vanishing-gradient problem that can occur with the sigmoid in deeper networks [4].

Regardless of the hidden layer activation, the output neuron applies a sigmoid to produce the predicted probability (*p-hat*) that a respondent belongs to the High Intention class. Both models are trained by minimizing the binary cross-entropy loss, defined in equation as below :

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)]$$

where N is the number of training samples, $y_i \in \{0,1\}$ is the true class label, and $(\hat{p})_i$ is the predicted probability. Both models use the Adam optimizer with an initial learning rate of 0.001 and are trained for a maximum of 1000 iterations. Input features are standardized to zero mean and unit variance prior to training to prevent dominance by features with larger scales.

3. Method

3.1. Data Collection

Primary data were collected via an online questionnaire distributed using Google Forms to students of Prasetya Mulya University between May 14 and May 17, 2026. A total of 300 valid responses were obtained after data cleaning. The survey instrument consisted of 18 closed-ended items measured on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree), grouped into six constructs: five predictor constructs ($X1-X5$) and one target construct (Y).

3.2. Variables and Operational Definitions

The five predictor constructs and the target construct are operationalized as follows:

- Y: Purchase Intention = Stated intention to buy, prioritize, and recommend Crocs products in the near future.
- X1: Price = Perceived affordability and value for money of Crocs products.
- X2: Promotion = Exposure to and perceived attractiveness of Crocs advertising, discounts, and brand collaborations.
- X3: Quality = Perceived durability, comfort, and design variety of Crocs products.
- X4: Customer Review = Influence of online reviews, word of mouth and KOL/influencer endorsements about Crocs
- X5: Brand Awareness = Top of mind awareness, brand recognition and perceived popularity of Crocs

3.3. Data Preprocessing

After collection, each construct score was computed as the mean of its three indicator items, producing six aggregated variables. The dataset was inspected for missing values; no missing entries were found among the 300 responses.

The continuous Purchase Intention score was then converted into a binary target according to the rule below :

$$y_{class} = 1 \text{ if } Y \geq 4, \text{ else } 0$$

where Y is the mean Purchase Intention score on the 1-5 Likert scale. The threshold of 4 corresponds to the “agree” level and separates respondents who express positive purchase intention from those who don’t.

Predictor features were then standardized using the z-score normalization, as shown below :

$$x_y^* = \frac{(x_{ij} - \mu_j)}{\sigma_j}$$

where x_{ij} is the value of the j-th predictor for the i-th respondent. μ_j and σ_j are the mean and standard deviation of predictor j across the training set, and x_{ij}^* is the standardized value used as input to the neural network.

3.4. Model Training and Evaluation

Both ANN models were trained and evaluated using stratified 5-fold cross-validation (K=5, shuffle = True, random_state = 42). For each fold, the model was trained on 80% of the data and tested on the remaining 20%. Evaluation metrics include accuracy, precision, recall, F1-score, and the confusion matrix. To identify which predictors contribute most strongly to the prediction, permutation feature importance was computed on the best-performing model using 20 repetitions.

4. Results and Discussion

4.1. Descriptive Statistics

As shown in **Table 1** and **Figure 1**, all variables have mean values above 3, which means respondents generally gave positive responses to each variable. Among the variables, ‘X5_Brand Awareness’ has the highest mean value (4.1689), indicating that respondents are highly aware of the brand. This result is also reflected in **Figure 1**, where Brand Awareness appears as the variable

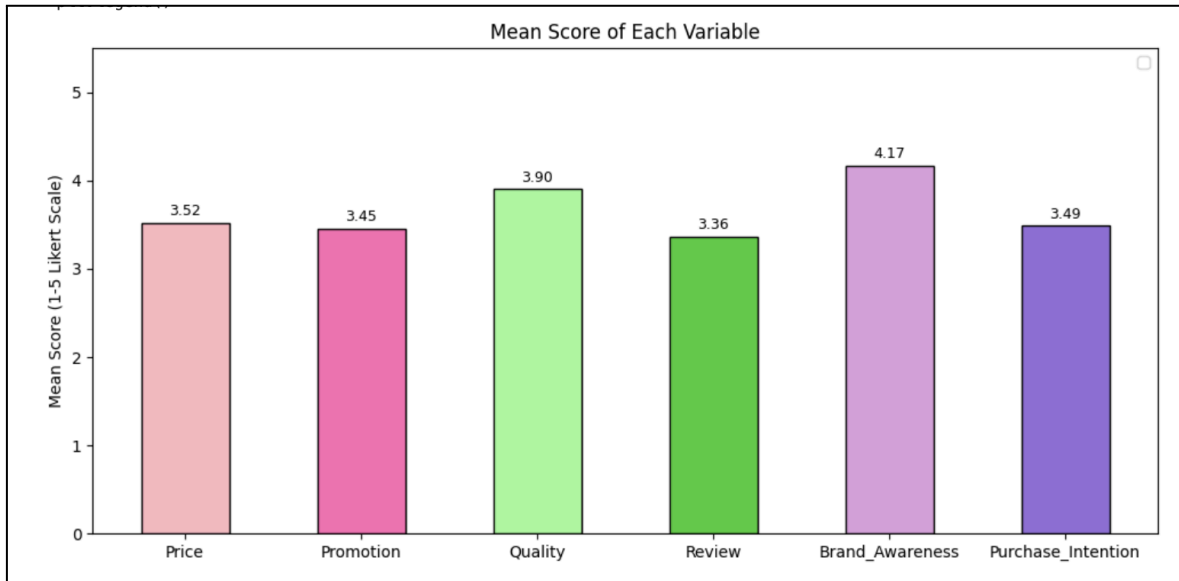
with the highest average score. In contrast, 'X4_Review' has the lowest mean value (3.3644), although the result still indicates a relatively positive perception among respondents.

The standard deviation values of all variables are below 1, indicating that the responses are fairly consistent and homogeneous. 'X2_Promotion' has the highest standard deviation value (0.8184), which shows that respondents have more varied opinions regarding promotional activities compared to the other variables. In addition, the minimum and maximum values range from 1 to 5 for all variables, indicating that respondents used the full range of the Likert scale in answering the questionnaire.

Table 1. Mean and Standard Deviation Table

Variables	Mean	Standard Deviation	Min	Max
X1_Price	3.5200	0.7300	1.0	5.0
X2_Promotion	3.4500	0.8184	1.0	5.0
X3_Quality	3.9033	0.6733	1.0	5.0
X4_Review	3.3644	0.8118	1.0	5.0
X5_Brand Awareness	4.1689	0.6916	1.0	5.0
Y_Purchase Intention	3.4867	0.8944	1.0	5.0

Figure 1. Variables' Mean Score

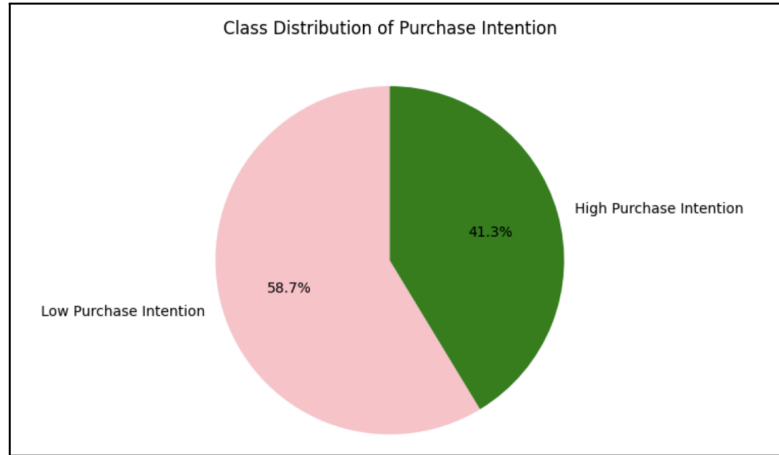


As shown in **Figure 2**, the distribution of purchase intention classes is relatively balanced, although respondents with low purchase intention have a slightly higher proportion. The graph shows that 58.7% of respondents are categorized as having low purchase intention, while 41.3% are classified as having high purchase intention.

This result shows that positive perceptions toward the variables such as brand awareness, quality, and promotion do not always lead to strong purchase intention among respondents. The difference between the two classes also suggests that there are still factors influencing consumers to hesitate before making a purchase decision. In the context of the ANN model, the class distribution

is still considered sufficiently balanced, allowing the model to learn patterns from both classes effectively and reduce prediction bias toward one category [8].

Figure 2. Class Distribution of Purchase Intention



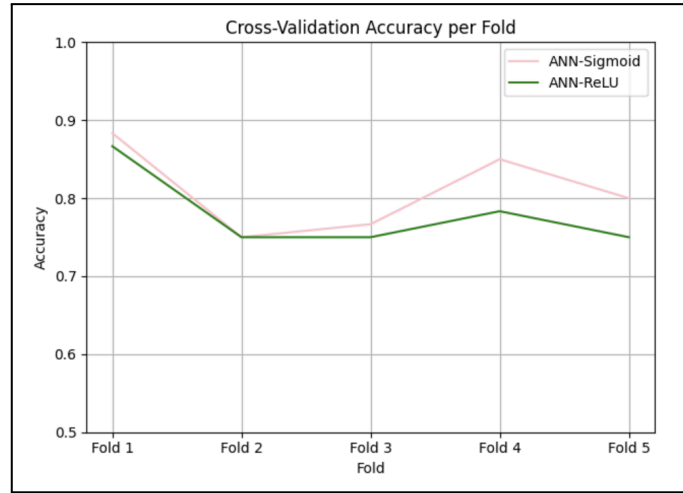
4.2. Cross Validation Results

As shown in **Table 2**, the Sigmoid activation function generally has higher accuracy compared to the ReLU activation function for all 5-fold cross-validation processes. The highest Sigmoid accuracy was obtained in 'Fold 1' with a value of 0.8833, while the highest ReLU accuracy was 0.8667 in the same fold. This trend can be seen in **Figure 3**, where the ANN-Sigmoid model consistently performed slightly better than the ANN-ReLU model in most folds. Both models experienced a decrease in accuracy in 'Fold 2' but the Sigmoid model showed better improvement in the following folds, especially in 'Fold 4'. In folds such as 'Fold 3', 'Fold 4', and 'Fold 5', Sigmoid also consistently outperformed ReLU.

Based on the average results, the Sigmoid model achieved a mean accuracy of 0.8100 ± 0.0501 , while the ReLU model obtained a mean accuracy of 0.7800 ± 0.0452 . This indicates that the Sigmoid activation function has slightly better classification performance for this dataset. The low standard deviation values for both models (± 0.0501 and ± 0.0452) also mean that the performance for all folds was fairly stable and consistent [9].

Table 2. Sigmoid vs ReLU Accuracy (5-Fold CV)

Fold	Sigmoid Accuracy	ReLU Accuracy
Fold 1	0.8833	0.8667
Fold 2	0.7500	0.7500
Fold 3	0.7667	0.7500
Fold 4	0.8500	0.7833
Fold 5	0.8000	0.7500
Mean Sigmoid Accuracy:		0.8100 ± 0.0501
Mean ReLU Accuracy:		0.7800 ± 0.0452

Figure 3. Cross-Validation Accuracy per Fold

4.3. Artificial Neural Network Model Evaluation

To further evaluate the performance of both ANN models beyond cross-validation accuracy, the classification reports for the ANN-Sigmoid and ANN-ReLU models are presented in **Table 3** and **Table 4**. These reports provide a comprehensive assessment using precision, recall, and F1-score, which are particularly informative given the moderate class imbalance in the dataset (176 low-intention vs. 124 high-intention cases).

Table 3 shows that the ANN-Sigmoid model achieved a balanced performance across both classes. For the ‘Low Intention’ class, it obtained a precision of 0.86, a recall of 0.81, and an F1-score of 0.83. For the ‘High Intention’ class, the model achieved a precision of 0.75, a recall of 0.81, and an F1-score of 0.78. The high recall for the high-intention class (0.81) indicates that the Sigmoid model is particularly effective at correctly identifying respondents who are likely to purchase Crocs, minimizing false negatives. The overall accuracy of 81% aligns with the cross-validation results, and the macro and weighted average F1-scores of 0.81 further confirm the model's consistency across both classes.

Table 3. ANN-Sigmoid Classification Report

	Precision	Recall	F1-Score	Support
Low Intention	0.86	0.81	0.83	176
High Intention	0.75	0.81	0.78	124
Accuracy			0.81	300
Macro Average	0.80	0.81	0.81	300
Weighted Average	0.81	0.81	0.81	300

In contrast, **Table 4** reveals that the ANN-ReLU model exhibited a noticeable imbalance in performance. While it achieved a respectable precision of 0.80 for the low-intention class and a high recall of 0.84, its performance on the high-intention class was weaker, with a recall of only 0.70. This relatively low recall suggests that the ReLU model misclassified a substantial number of high-intention respondents as low-intention (false negatives). Consequently, the F1-score for the

high-intention class was only 0.73, which is notably lower than that of the Sigmoid model. The macro average F1-score of 0.77 and the weighted average of 0.78 for the ReLU model are also inferior to those of the Sigmoid model.

Table 4. ANN-ReLU Classification Report

	Precision	Recall	F1-Score	Support
Low Intention	0.80	0.84	0.82	176
High Intention	0.75	0.70	0.72	124
Accuracy			0.78	300
Macro Average	0.77	0.77	0.77	300
Weighted Average	0.78	0.78	0.78	300

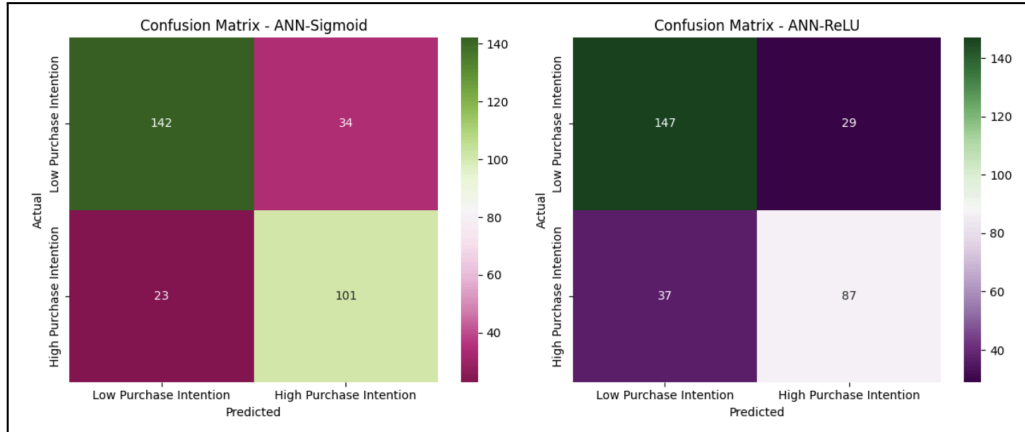
Table 5 provides a side-by-side comparison of the key evaluation metrics. The ANN-Sigmoid model consistently outperforms the ANN-ReLU model across all four metrics: accuracy (0.81 vs. 0.78), precision (0.748 vs. 0.75), recall (0.815 vs. 0.702), and F1-score (0.780 vs. 0.725). The most substantial difference is observed in recall, where the Sigmoid model demonstrates a clear advantage in identifying high-purchase-intention respondents. This finding is consistent with the observation made by Glorot et al. [10], who noted that while ReLU is highly efficient in deep networks due to its ability to mitigate the vanishing gradient problem, its performance can be suboptimal in shallow architectures or with smaller datasets, where the bounded nature of the sigmoid function may still be beneficial.

Table 5. Model Evaluation Results

Model	Accuracy	Precision	Recall	F1-Score
ANN-Sigmoid	0.81	0.748148	0.814516	0.779923
ANN-ReLU	0.78	0.75	0.701613	0.725

As shown in **Figure 4**, the ANN-Sigmoid model correctly classified 142 respondents with low purchase intention and 101 respondents with high purchase intention. Meanwhile, the ANN-ReLU model correctly classified 147 low purchase intention cases and 87 high purchase intention cases. Although the ANN-ReLU model performed slightly better in identifying low purchase intention respondents, it produced more misclassifications in the high purchase intention category. On the other hand, the ANN-Sigmoid model showed more balanced classification results with fewer prediction errors.

Overall, the ANN-Sigmoid model can be considered the better model for this study since it has higher accuracy and more balanced performance for both purchase intention categories.

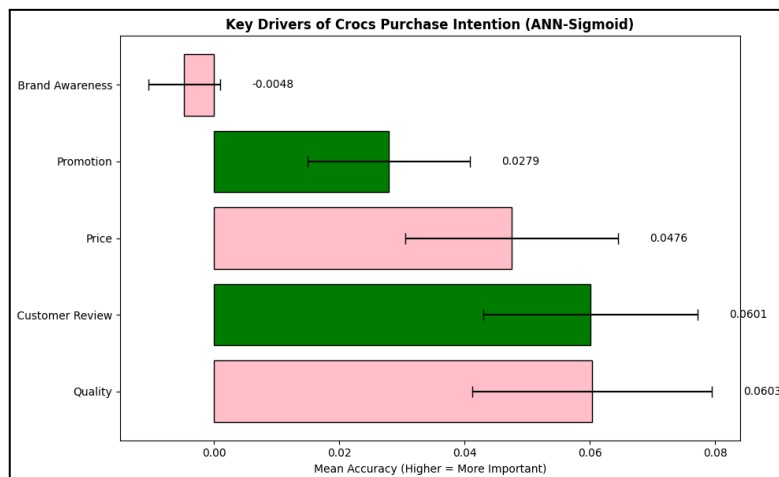
Figure 4. Cross-Validation Accuracy per Fold

4.4. Best Features

To determine which predictors most strongly influence Crocs' purchase intention, permutation feature importance was calculated using the best-performing model (ANN-Sigmoid). As presented in **Table 6** and visualized in **Figure 5**, Quality (mean importance = 0.0603) and Customer Review (0.0601) emerged as the top two drivers, with nearly identical impact. Price ranked third (0.0476), followed by Promotion (0.0279). While, Brand Awareness showed a negative importance (−0.0048), indicating that this feature does not help distinguish between high and low intention respondents.

Table 6. Feature Importance

	Feature	Importance	Std
0	Quality	0.060333	0.019155
1	Customer Review	0.060111	0.017137
2	Price	0.047556	0.016995
3	Promotion	0.027889	0.012981
4	Brand Awareness	-0.004778	0.005753

Figure 5. Permutation Feature Importance of ANN-Sigmoid Model

Kotler and Keller [5] state that "brand awareness is necessary but not sufficient to drive purchase intention; consumers must also perceive superior product quality and value." In this study, although respondents demonstrated high brand awareness (mean = 4.17), this factor alone did not translate into purchase intention. Instead, the model consistently identified product quality and customer reviews as the strongest predictors. This finding supports the argument that for consumers with existing brand familiarity, tangible product attributes and social proof play a more decisive role than mere brand recognition.

Therefore, for the Indonesian university student segment, Crocs should prioritize product quality improvements, actively manage online customer reviews, and maintain competitive pricing strategies over brand visibility campaigns alone.

5. Conclusion

This study classified Crocs purchase intention among 300 Prasetya Mulya University students using an Artificial Neural Network with two different activation functions. The Sigmoid model achieved a mean 5-fold cross validation accuracy of 81% and an F1-score of 0.78, outperforming the ReLU model. Permutation feature importance further revealed that Quality, Customer Review, and Price are the strongest drivers of Crocs purchase intention, while Brand Awareness has a negligible effect.

Future research can extend this work in several directions. First, larger and diverse samples beyond a single university would improve the external validity. Second, additional activation functions (such as TanH, Leaky ReLU, and ELU) and other architectures could be benchmarked. Third, incorporating psychographic or behavioral variables (such as fashion, social influence) and other variables could further improve the predictive performance. Lastly, the model can be deployed as a decision support tool for marketers to segment consumers based on predicted purchase intention.

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